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Two-dimensional (2D) Mesoscale self-assembly (MESA) experiments, conducted in the Whitesides group at Harvard, have led to the biomimetic self-assembly of mesoscale DNA analogs and folding of mesoscale amino acid analogs. The fact that mesostructures can be directly observed allows visualization of the dynamics of self-assembly. Thus, MESA experiments provide important experimental data for computational studies of self-assembly, such as protein folding algorithms. In addition, 3D MESA experiments have produced functional 3D electrical networks that have self-assembled by simply mixing the components. Moreover, these electrical networks could be assembled in series or in parallel by changing the recognition site patterning. These MESA experiments share important principles with biological self-assembly, including shape-selective recognition and noncovalent binding. These principles enable self-assembly of complex architectures that are inherently self-healing due to the reversible nature of noncovalent binding and the continuous sampling of configurational space to reach the structure representing the thermodynamic

	minimum. Here we propose a four-year ground-based instrument development program that will enable the study of MESA in microgravity. The initial ground-based experiments will provide the prerequisite knowledge and experience base for future microgravity experiments. For 3D microgravity MESA, a mixing chamber will be developed and constructed. The microgravity environment eliminates the need for an isodense medium used in 1 g experiments to suspend 3D meso objects, and allows isotropic mixing. For 2D variable gravity MESA, existing ground-based and flight microscope centrifuges can be slightly modified to satisfy experimental requirements. Machine vision algorithms will be implemented for automated analysis of MESA product distribution. Variable gravity 2D MESA experiments will enable continuous control of the bond strength and range of interaction between MESA objects. These parameters govern the size limit of MESA objects, dynamics of self-assembly, and distribution of self-assembled products. Thus, the proposed experiments will provide the fundamental knowledge needed to build functional self-assembled products such as self-assembled and self-repairing 3D microelectronics.
Task Description:	The Whitesides group at Harvard has been exploring capillary force driven self-assembly of mesoscale (nm to mm) objects into regular arrays. The self-assembling systems discussed here are divided into two categories: (1) the self-assembly of polymeric objects at the water/perfluorodecalin (H2O/PFD) interface into 2D structures, and (2) the self-assembly of polymeric objects into 3-D arrays. The 2D work has produced biomimetic analogs of protein folding and DNA hybridization , and the 3D work has produced the first self-assembly of functional 3D electrical networks. In a collaborative effort, the JPL-based group would like to extend this research into a microgravity environment. In 2D MESA, the microgravity environment enables us to study the nature of the capillary bond by tuning the bond strength and range of interaction between mesoscale objects using variable gravitational force. Computations show that a decrease in gravitational force will lead to an increase in bond strength, thus permitting construction of smaller microstructures. Moreover, changing the bond strength and range of interaction will shift the equilibrium between possible mesostructure architectures, thus providing a handle for controlling the product distribution of self-assembly. We can tune the gravitational force, using existing microscope centrifuge technology from ~ 0 g to 1.5 g in microgravity, and 1 g to 10 g in ground-based experiments. In addition, the microgravity environment enables novel assembly strategies for 3D microstructures. For example, simply changing the humidity in the mixing chamber will induce crystallization of the free-floating objects. High humidity wets the surfaces of the objects, which gives rise to the attractive capillary force. Here we propose the development of a ground-based and flight microscope centrifuge and subsequent variable gravity 2D MESA experiments. Mesoscale self-assembly (MESA) methods will provide new routes to materials and functional systems that cannot be fabricated by other means. Beyond the simple self-assembled circuits already demonstrated, MESA has the potential to generate complex functional structures such as microelectronics, photonic band gap materials, waveguides, high surface area catalysts, and microelectromechanical systems (MEMS). Ultimately, the utility of MESA will depend on how small the components can be made and what interesting structures can be assembled. To this end, understanding the physics of capillary bond formation will lay the foundation for design of intricate functional systems. This proposal outlines the fundamental studies that will test the limits of capillary-bond mediated self-assembly of microstructures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lessons learned from mesoscale self-assembly may ultimately lead to new manufacturing processes of electronics components.
	A: Project Objectives (For Year 3) - Perform variable gravity mesoscale self assembly experiments. - Analyze meniscus shape as a function of centrifugal force. B: Introduction: Mesoscale Self-Assembly (MESA) is an unaided process in which objects subjected to random motion find the thermodynamic minimum, self-assembling into ordered arrays through non-covalent forces. In physics, “meso” (middle) refers to an object with dimensions comparable to those of the phenomena under investigation. Current work in MESA explores the force that drives self-assembly, the capillary bond. It follows a roughly exponential curve, and is approximately on the same scale as the height of the face; thus, the scales involved in current research on self-assembly range from nanometers to centimeters. MESA naturally lends itself to being a biomimetic model, since life itself is essentially self-assembled, on a molecular level. Processes, such as DNA hybridization and enzyme–substrate association, and parts, such as the human brain, are all a result of molecular self-assembly. MESA shares many fundamental principles with molecular self-assembly, and therefore is a convenient modeling mechanism, since MESA is large enough to observe and manipulate with minimal effort. In both MESA and molecular self-assembly, a single, desired complex stably forms, despite many possible configurations. Such specificity is due to the reversible nature of noncovalent binding, which allows continuous sampling of configurational space until selective recognition creates a product that can survive at the thermal energy of the system. As a result, the product is not only stable, but also self-healing: should the product be broken up again, it will repair itself, and spontaneously reform into the desired configuration. MESA is being explored on both the 2D and 3D level. Both 2D and 3D MESA usually occurs at a fluid-fluid interface. Two-dimensional MESA typically involves a flat fluid-fluid interface, like a layer of water sitting on a layer of oil. Currently, three-dimensional MESA takes place on a drop of an immiscible liquid suspended in an isodense liquid, generating a three-dimensionally shaped interface, similar to 2D MESA. The shape of the drop can be manipulated, allowing the formation of spheres, cones, and other 3D shapes. Research into the 3D field can potentially yield 3D self-fixing computers which function like brains. Current work includes the formation of simple circuits and submillimeter-sized spheres from hexagonal pieces, which, once welded together, were strong enough to be handled in air. Two-dimensional MESA yields modeling of simple biological processes such as enzyme/substrate activation, DNA hybridization, and polymerization. In this project, we are investigating 2D MESA, focusing on the capillary forces, which largely determine the binding pattern of the MESA pieces. This includes studying the effect of gravity on the meniscus of a piece of PDMS sitting at a water-perfluorodecalin (PFD) interface. PFD is hydrophobic, and immiscible (not mixable) in water. Typically, the sides of the pieces of PDMS are patterned to be either hydrophilic or hydrophobic, depending on the

desired result. The sides of the PDMS either pulls up or pulls down the water-PFD interface, creating either a positive (hydrophobic side) or negative (hydrophilic side) meniscus. To minimize the interfacial free energy, like sides attract through capillary forces, and opposites repel. The dominant force is determined by the size of the meniscus: a denser piece floats lower in the meniscus, which makes the negative meniscus (hydrophilic sides) stronger than the positive one. In other words, the hydrophobic sides will still attract, but binding will favor the hydrophilic sides – the hydrophobic sides' attractive forces may not be enough to overcome the thermal energy of the environment, so that only the hydrophilic sides will create a stable bond.

For two-dimensional MESA, the disruptive force that causes selective binding is orbital oscillation. The oscillation rate affects which configurations survive: higher rates allow only short sections, or stable circles to survive, which lower rates give rise to long, branched sections.

Calculations indicate that a decrease in gravitational force leads to an increase in bond strength, thus allowing greater control over the product distribution of self-assembly. Using a centrifuge, we measured the properties of the meniscus as the gravitational force applied changed, and determined the effect of the change of gravity on the self-assembly process by analyzing the change of the shape of the meniscus.

C. Methods:

Materials: The MESA pieces used for this experiment were fabricated from polydimethylsiloxane (PDMS), and hand-placed at the water-perfluorodecalin (PFD) interface. PDMS was chosen as the material for the MESA pieces because it is easy to dope, mold into various shapes, fabricate, and is relatively inexpensive.

The water-PFD interface was chosen because it does not cause PDMS to swell, and the density of water and of PFD conveniently brackets that of PDMS, though the density of both water and PDMS can be easily altered. Also, PFD is a nonpolar, hydrophobic liquid, and immiscible in water. The water-PFD combination provides high interfacial free energy, which creates the strong capillary forces needed to run MESA. It is this interface that is the subject of study.

The gelatin used to solidify the water layer came from porcine skin, was Type A, had strength 300 bloom, and was ordered from Sigma-Aldrich. The casting material was Norland Optical Adhesive 68 (NOA 68).

MESA Piece Fabrication: The pieces used in MESA were fabricated from polydimethylsiloxane (PDMS), which is made by mixing silicone elastomer base and curing agent in a 10:1 ratio by mass. PDMS is naturally hydrophobic, and has a density $\rho = 1.05$ g/cm3. Doping PDMS can change its properties. Possible doping agents include Al2O3, which has a density $\rho = 4$ g/cm3 and can be used to adjust the density of a piece of PDMS according to the formula:

D = desired density Q = grams pure PDMS to be doped A = grams Al2O3 needed to achieve D

The pieces can also be doped with fluorescing powder, which, under UV light, makes the pieces easy to distinguish on videotape. Adding the powder also lessens blurring on video recordings. Since a very small amount of fluorescing powder is, the density of each piece is considered to be unaffected by that addition. However, for simplicity, and a maximally tall meniscus, this experiment used undoped PDMS. PDMS molds are made by filling a tube with undoped, degassed PDMS, and inserting a hexagonal rod coated with (tridecafluoro-1,1,2,2-tetrahydrooctyl)-1-trichlorosilane. This is placed in an oven at 60°C – 80°C for 1-2 hours. As the PDMS cures, it conforms to and preserves the shape of the rod. After curing, the rod is removed, and the mold is placed in a vacuum desiccator with an open container of (tridecafluoro-1,1,2,2-tetrahydrooctyl)-1-trichlorosilane. The chamber is evacuated, and left overnight to silanize the molds.

The next day, after mixing, more PDMS is degassed in a vacuum desiccator, to remove any air bubbles which could change the density of the pieces. The degassed PDMS is then carefully poured into the PDMS mold, whose insides are now coated with (tridecafluoro-1,1,2,2-tetrahydrooctyl)-1-trichlorosilane or a similar substance, which prevents the uncured PDMS from bonding to the mold during curing. The PDMS is then left in an oven to cure at 60°C – 80°C for 1-2 hours.

After the PDMS has cured in its mold, the mold is removed from the oven, and left to cool. Then the PDMS rod or shape is carefully removed from the mold.

The PDMS rod can then be patterned, by taping over the sides to be doped hydrophilic. Then, the rod is placed in a Petri dish, and again covered with uncured PDMS to form a protective layer. After curing this, the tape is carefully removed, revealing the sides of the rod to be doped hydrophilic. Then the rod is cut into slices twice the desired thickness of each piece, and placed in a plasma oxidizer for a few minutes. After oxidation, the pieces are removed from the oxidizer, marked, and peeled of the remaining protective layer of PDMS. Each piece is then cut in half, so that every piece has one hydrophilic and one hydrophobic face, and patterned sides.

Task Progress:

The individual pieces are carefully sliced to uniform thickness using a guillotine, pictured above. Then, they are placed at the water-PFD interface. It is important to insert the pieces carefully, to avoid creating bubbles which can interfere with images of the meniscus.

Improvements to Experiment: Imaging the meniscus is a difficult problem. Previous attempts by the Whitesides group, using Nomarski-mode spectroscopy, laser beam reflection from the interface, and diffraction based-designs have all been unsuccessful.

While this setup has the potential to capture the most accurate image of the meniscus, it has proven to be neither the most efficient nor most realistic solution for several reasons.

First of all, there is the issue of finding a remote/wireless camera with a very short focal length that is robust enough to withstand the forces it would be subject to in the centrifuge, yet small enough to fit into the setup. Such a camera could not be found. Even if such a camera had been found, the designing of the setup such that it could realistically be machined is another non-trivial issue. The entire setup must be fully constrained, yet sit statically in a 250 mL centrifuge bottle with a tapered opening. Additionally, the setup should be adjustable, so that if some part of it has unexpected alignment issues, the entire setup does not have to be re-built. The piece of PDMS must be completely static with respect to the rest of the system, and rest exactly at the interface. Building something to constrain the piece of PDMS would be difficult: for reference, the cuvette has a volume of only 3 mL, yet is approximately two inches high. The entire setup must also be illuminated, bright enough to see the meniscus clearly, yet without glare.

After reading through several papers on MESA, a new method was discovered, first proposed by the Whitesides Group at Harvard. They solidified the water layer of the interface by adding gelatin to the water layer, and cooling the system until the gelatin set. After that, the gelatin was removed from the container, the PDMS tile removed from the gelatin, and the gelatin's interface-facing bottom dried with N2 from a hose. Then, a layer of Norland Optical Adhesive 68 (NOA 68) was spread over the indentation in the gelatin made from the PDMS tile, and a bit of the surrounding area, and cured under an ultraviolet lamp for about an hour. Once cured, NOA 68 can be handled, and is an exact replica of the PFD layer. The NOA 68 tile was then sliced and photographed.

This method has been proven to faithfully recreate the pattern imprinted on the gelatin to the micron level. While the addition of gelatin to the system slightly lowers the interfacial free energy of the PFD/water interface, this is a comparative study, and since the level of gelatin was held constant across all trials, this method was chosen. Also, a lamp stand was machined to securely hold the UV lamp during the curing process.

Procedure: Add approximately 70 mL of PFD, then 70 mL of water to each centrifuge tube, pouring carefully to avoid bubbles. Then heat 70 mL of water more for each tube, and add 6.0 g gelatin to the heated water. Stir the hot water until the gelatin completely dissolves. Slowly add the gelatin solution to each centrifuge tube. After that, weigh each centrifuge tube, and add water until the weight of each tube is within .05 g of the other. Place the centrifuge tubes in the adaptors, and place the adaptors in the centrifuge rotor. Set the centrifuge to the desired RPM (rotations per minute) or RCF (relative centrifugal force), temperature, and time (1.5 hours), and start the centrifuge. RPM and RCF are related through the following equation:

RCF = 176.96*(RPM/1000)2.

Allow the system a minute to come to equilibrium, then cool the system to 4°C. Allow the cooled centrifuge to run for 1.5 hours. After this, remove the tubes from the centrifuge, and cut out the section of gelatin with the PDMS tile. Remove the PDMS piece, and dry the interface surface of the gelatin with nitrogen for 10 seconds. Then add NOA 68, and cure for an additional hour. Remove the NOA 68 tile from the gelatin, and carefully cut a thin slice, perpendicular to the side of the indentation created by the PDMS piece. After this, capture an image of the slice. After obtaining a micrograph, clean the image of the background and dust in Photoshop, and save the image as a grayscale jpeg. Since there is no absolute way to align the pictures that takes all problems such as tilting into account, the images for this experiment were aligned by attempting to match the PDMS side of the graphs, and the flat, constant part of each meniscus. In this way, the images of the meniscus in 1g provide a basis with which to compare the other menisci. After this, analyze the picture in MATLAB and fit an exponential equation to the curve of the meniscus of the form c*e-ax. Lastly, tabulate the results.

D Results: The meniscus was imaged at 1g in a beaker, 1g in the centrifuge tube, 100 RPM, 200 RPM, and 5,300 RPM. The menisci curves were analyzed and fitted exponentially in Matlab to yield the following results: y=c*e-ax where c=eb, a constant

DATA

The data shows that the values of a and b are not affected by the centrifuge, and that experiments done in a beaker will yield the same results as experiments done in the centrifuge at the same gravitational force. The data also indicates a trend, in which the value of a increases with increased gravity, and the value of b correspondingly decreases. This is what one would intuitively expect: as gravity exerts greater force on the meniscus, the exponential curve decays faster, resulting in a larger curve in the meniscus (a), and a shorter overall height (b).

All menisci pictures were aligned before cropping the pictures at 2mm from the edge of the PDMS piece.

These pictures support the data table above: It is clear that the curvature of the meniscus increases with gravity, while the overall height correspondingly decreases.

E. Conclusion: The above data implies a direct correlation between gravitational force and meniscus shape, and confirms the theory that the capillary force and range of interaction is inversely related to gravitational force. Thus, as gravitional force is decreased in microgravity, we expect that mesoscale particles will bind much more strongly to each other and exhibit attractive and repulsive interaction over longer distances with respect to 1 G.

F. Future Work: It would have been interesting to have made a quantitative analysis of the effect of gelatin on the meniscus. However, to image the meniscus, and accurately calibrate the pictures would require more equipment, and possibly a different setup.

It would also be interesting to run this experiment in microgravity, from ~0g to 1g, to see if the results interpolated from this experiment holds true over the entire range of gravitational force, or where it breaks down.

Future work on this project will involve repeating this experiment more rigorously, performing perhaps five to ten trials per gravitational level. It would also include devising a method that would allow one to run MESA experiments in the centrifuge, so that the product distribution at each gravitational level can be analyzed, and the results of the findings from the meniscus analysis can be either confirmed or deemed to not be true-to-life.

C: Significance of Results:

Past work in mesoscale self-assembly (MESA) has produced simple self-assembling circuits, and directly observable biomimetic analogs of protein folding and DNA hybridization. Due to the reversible nature of the capillary bond, MESA has the potential to create complex functional structures, such as microelectronics and microelectromechanical systems, that are not only self-assembling, but also self-healing. The strength of the capillary bond between pieces of polydimethylsiloxane (PDMS) can be determined by analyzing the energy of the water-perfluorodecalin interface at which the pieces of PDMS float. By taking a cast of the system, one can obtain an accurate image of and therefore fit a function to the meniscus, obtaining a greater understanding of the physics of capillary bond formation and laying the foundation for the design of intricate functional systems. Calculations indicate that a decrease in gravitational force leads to an increase in bond strength, thus allowing greater control over the product distribution of self-assembly. Using a centrifuge, we measured the properties of the meniscus as the gravitational force changed, and determined the effect of the change of gravity on the self-assembly process.

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